AGE CALCULATOR

C PROGRAMMING MINI PROJECT

PROJECT DOCUMENTATION

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS OF THE DEGREE OF

DIPLOMA IN

COMPUTER ENGINEERING

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UNDER THE GUIDANCE OF

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2026 – 2027

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CERTIFICATE

This is to certify that,

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Have satisfactorily carried out the C PROGRAMMING MINI PROJECT work titled "***Age -Calculator***" in partial fulfillment of the degree of Bachelor of Engineering as laid down by the University of Mumbai during the academic year 2026-2027.

Guide Name

Supervisor / Guide

Mr. Nitish Jha

DECLARATION

I hereby declare that the mini project titled “Age Calculator” has been developed by me as part of the C Programming coursework. The work presented in this report is based on the implementation and testing carried out for this project. Any external references used are listed in the bibliography.

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Chapter 1 – Introduction

1.1 Project Overview

Age Calculator is a console-based mini project developed using the C programming language. It accepts the user's date of birth and calculates the age in years, months and days using the computer's current system date.

1.2 Problem Statement

Calculating an exact age in years, months and days manually can be inconvenient because month lengths and leap years differ. This project provides a simple program to perform the calculation and validate the entered date.

1.3 Objectives of the Project

- Accept date of birth in DD MM YYYY format.
- Obtain the current date from the system clock.
- Calculate age in years, months and days.
- Validate calendar dates, including leap years.
- Reject invalid input and future dates.

1.4 Scope of the Project

The current version is a single-user console application. It calculates age for one date of birth per execution. It does not store personal details, calculate age for multiple users in a database, or provide a graphical/web interface.

Chapter 2 – Requirement Analysis

2.1 Hardware Requirements

A smartphone, tablet or computer with an internet connection and a modern web browser is sufficient when using the OnlineGDB online compiler. A keyboard or on-screen keyboard is required to enter the date.

2.2 Software Requirements

- Programming language: C
- Online compiler: OnlineGDB C Compiler (https://www.onlinegdb.com/online_c_compiler)
- Web browser with internet access
- GitHub account and web interface for source-code hosting

No local compiler installation is required for the online development workflow used in this project.

Chapter 3 – System Design

3.1 System Architecture / Block Diagram

The program follows a simple input–process–output structure:

10. User enters date of birth (day, month, year).
11. Input validation checks the date and rejects future dates.
12. The system date is obtained using the time library.

13. Age is calculated by subtracting the birth date from the current date, borrowing days/months when required.
14. The calculated age is displayed in years, months and days.

Block diagram (text representation):

User Input → Date Validation → Get Current Date → Age Calculation → Display Result

3.2 Algorithm

15. Start the program.
16. Read day, month and year of birth.
17. Get the current system date.
18. Check whether the entered date is valid.
19. Check that the date of birth is not in the future.
20. Subtract dates and adjust negative day/month values using the previous month's length.
21. Display age in years, months and days.
22. Stop.

Chapter 4 – Implementation

4.1 Technologies Used

The project is written in C. It uses structures, functions, conditional statements, arrays, input/output operations and the time library. The header files used are stdio.h and time.h.

4.2 Source Code

The complete source code is provided below and is saved in the GitHub repository as main.c.

```
#include <stdio.h>
#include <time.h>

struct Date {
    int day;
    int month;
    int year;
};

int isLeapYear(int year) {
    return (year % 400 == 0 ||
            (year % 4 == 0 && year % 100 != 0));
}

int daysInMonth(int month, int year) {
    int days[] = {
        31, 28, 31, 30, 31, 30,
        31, 31, 30, 31, 30, 31
    };

    if (month == 2 && isLeapYear(year))
        return 29;

    return days[month - 1];
}

int isValidDate(struct Date d) {
    if (d.year < 1 || d.month < 1 ||
        d.month > 12 || d.day < 1)
        return 0;
```

```
    return d.day <= daysInMonth(d.month, d.year);
}

int main() {
    struct Date dob, today, age;
    time_t now = time(NULL);
    struct tm *current = localtime(&now);

    if (current == NULL) {
        printf("Unable to get current date.\n");
        return 1;
    }

    today.day = current->tm_mday;
    today.month = current->tm_mon + 1;
    today.year = current->tm_year + 1900;

    printf("===== AGE CALCULATOR =====\n");
    printf("Enter your Date of Birth (DD MM YYYY): ");

    if (scanf("%d %d %d",
        &dob.day, &dob.month, &dob.year) != 3) {
        printf("Invalid input!\n");
        return 1;
    }

    if (!isValidDate(dob)) {
        printf("Invalid date of birth!\n");
        return 1;
    }

    if (dob.year > today.year ||
        (dob.year == today.year &&
         dob.month > today.month) ||
        (dob.year == today.year &&
         dob.month == today.month &&
         dob.day > today.day)) {
        printf("Date of birth cannot be in the future!\n");
        return 1;
    }

    age.year = today.year - dob.year;
    age.month = today.month - dob.month;
    age.day = today.day - dob.day;

    if (age.day < 0) {
        age.month--;
        int previousMonth = today.month - 1;
        int previousYear = today.year;

        if (previousMonth == 0) {
            previousMonth = 12;
            previousYear--;
        }

        age.day += daysInMonth(previousMonth, previousYear);
    }

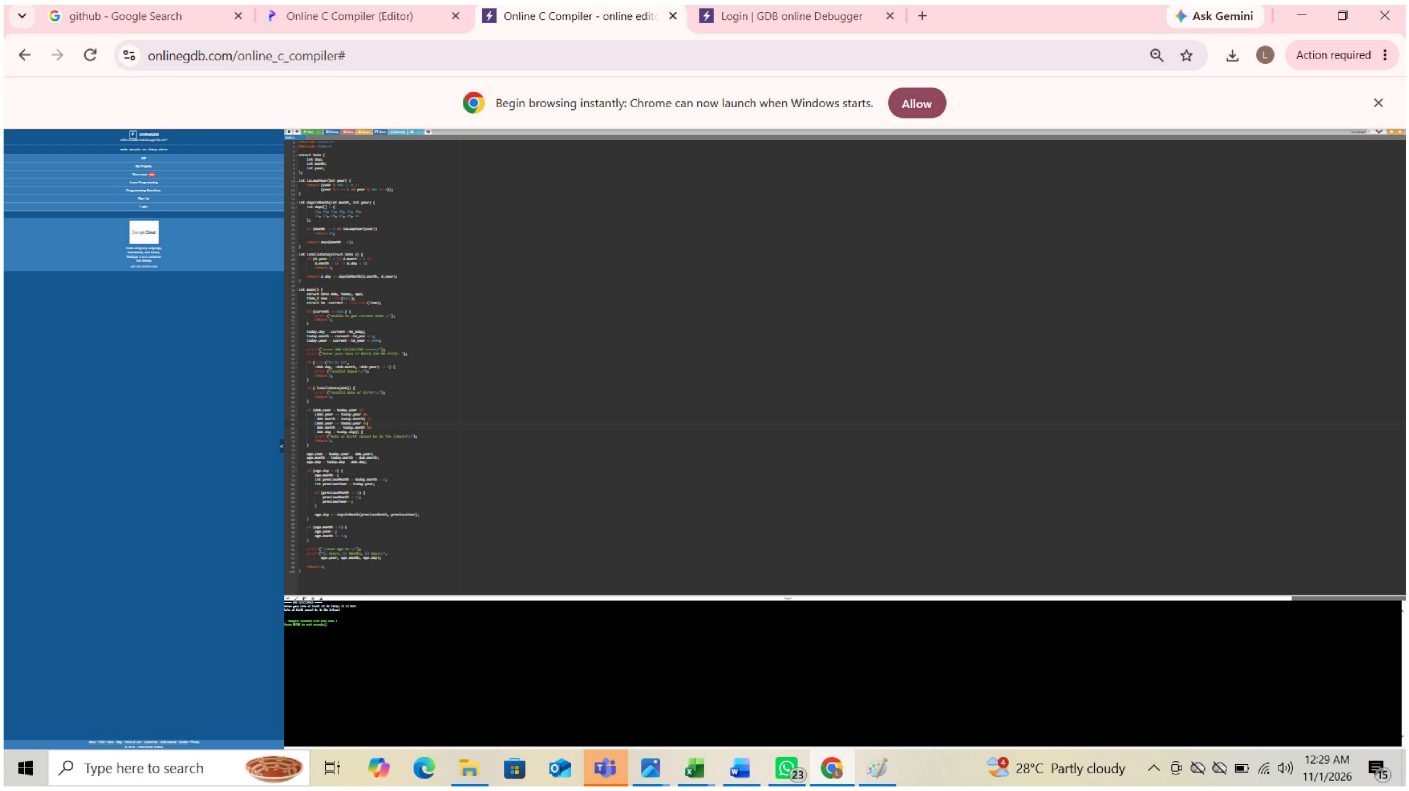
    if (age.month < 0) {
        age.year--;
        age.month += 12;
    }

    printf("\nYour Age is:\n");
    printf("%d Years, %d Months, %d Days\n",
           age.year, age.month, age.day);

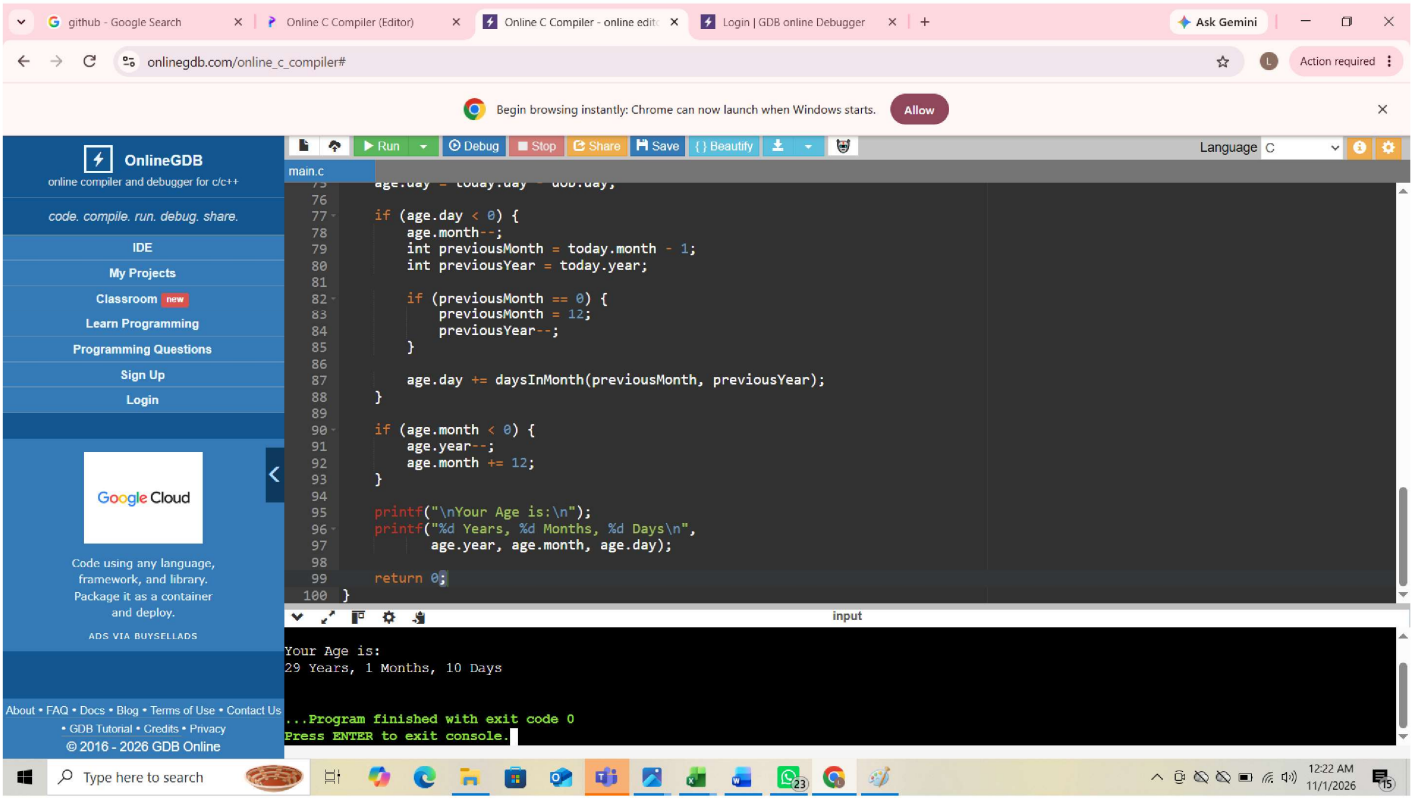
    return 0;
}
```

4.3 Screenshots of Implementation

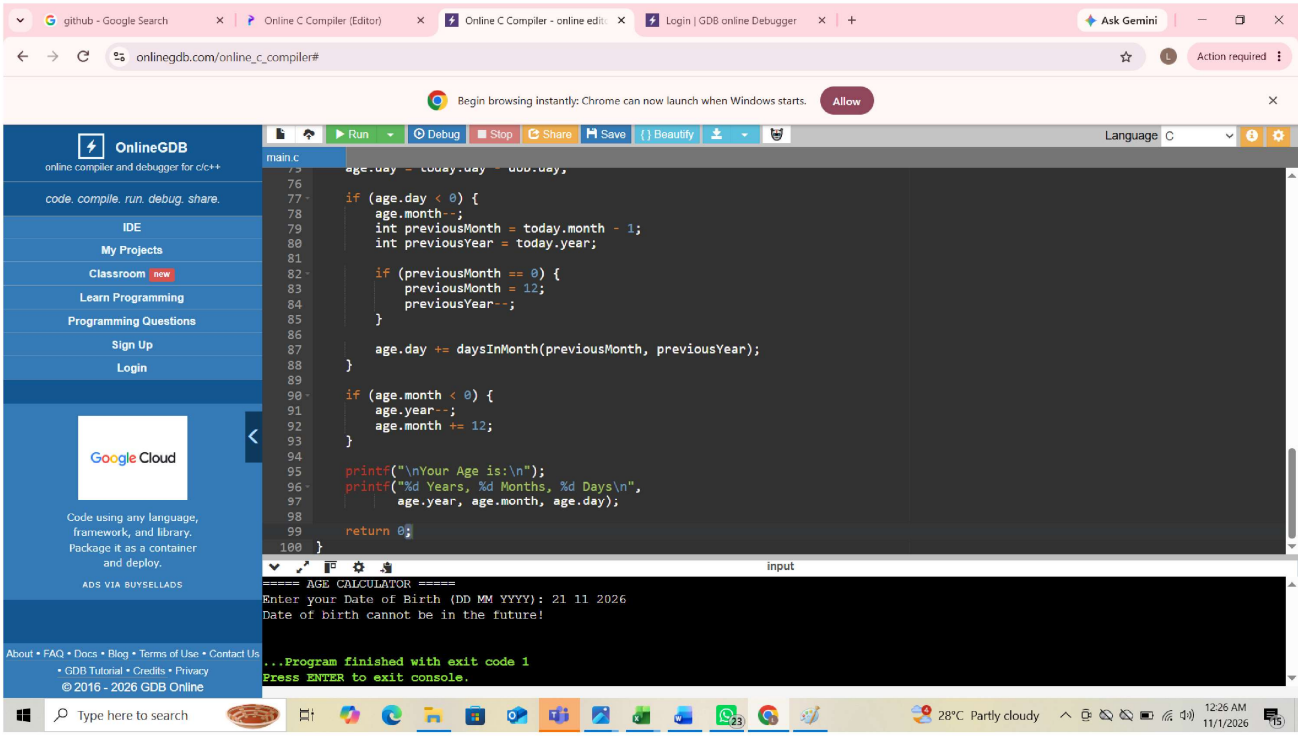
1. Full code with output



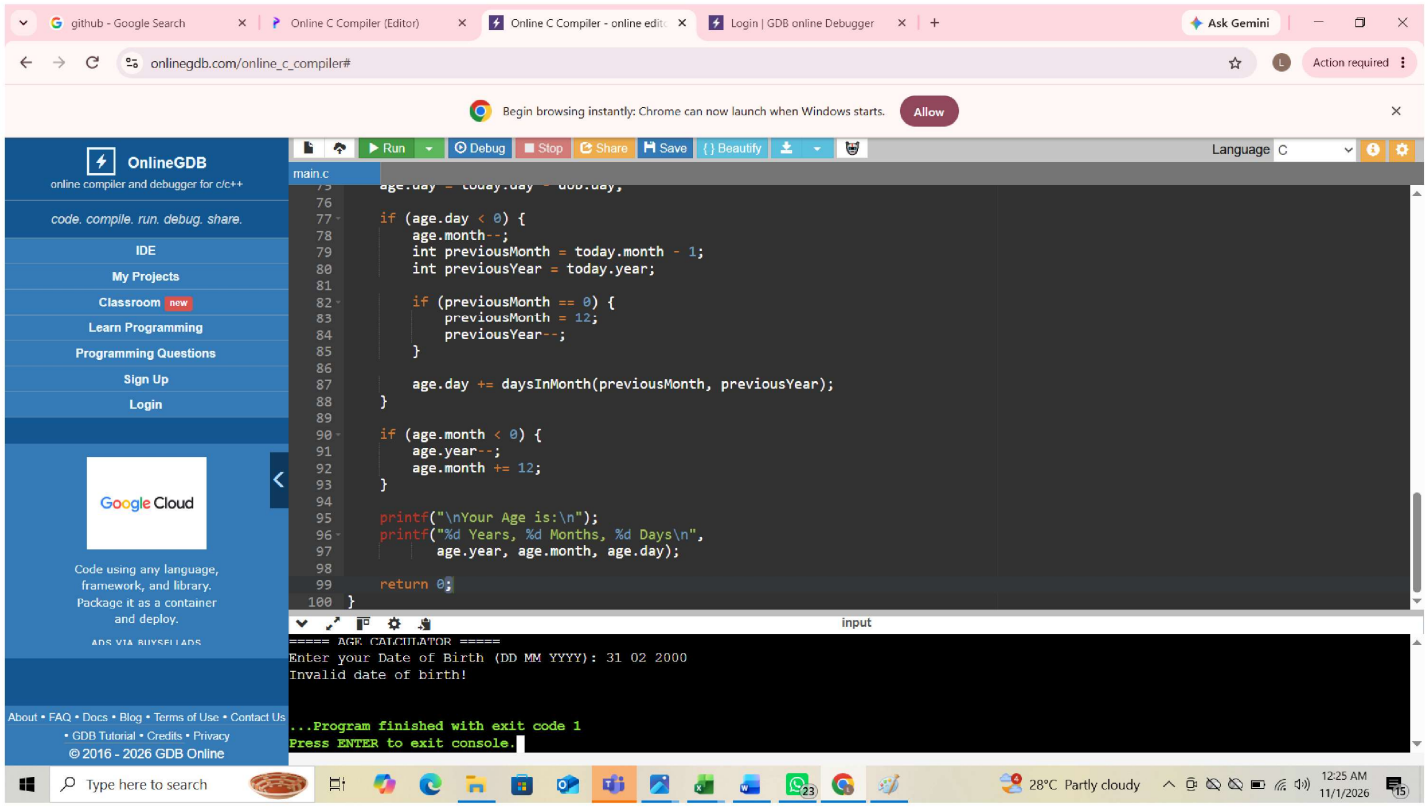
2. Out put



3. Future Date Error



4. Invalid Date Error



Chapter 5 – Project Output and Testing

5.1 Execution of the Project

Open the OnlineGDB C compiler, select C as the language, paste the source code into the editor and press Run. When prompted, enter the date of birth as three integers separated by spaces in DD MM YYYY format.

5.2 Explanation of Project Output

For a valid date of birth, the program displays the calculated age in years, months and days. For invalid dates, non-numeric input or a future date, it displays an error message and terminates.

5.3 Test Cases

Test	Input / Condition	Expected Result
1	16 03 2011 (assuming current date 01-11-2026)	26 Years, 2 Months, 17 Days
2	31 02 2000	Invalid date of birth!
3	A date later than the current system date	Date of birth cannot be in the future!
4	Non-numeric input	Invalid input!

Note: Run each test in the online compiler and insert actual screenshots. The displayed current date depends on the compiler's system date/time.

Chapter 6 – GitHub Repository and Version Control

6.1 Introduction to Git and GitHub

Git is a version control system used to track changes in source files. GitHub is an online platform for hosting Git repositories and sharing project source code.

6.2 Creating a GitHub Repository

A public repository named Age-Calculator was created under the GitHub account lavanya-agecalculator.

6.3 Uploading Project Source Code to GitHub

The file main.c and README.md were added to the repository through the GitHub website and saved using the Commit changes option.

6.4 Git Commands Used

The project was uploaded using the GitHub web interface; command-line Git was not used in the described workflow. The following are standard commands for a future local Git workflow:

```
git init
git add .
git commit -m "Initial project commit"
git branch -M main
git remote add origin <repository-url>
git push -u origin main
```

6.5 GitHub Repository Link

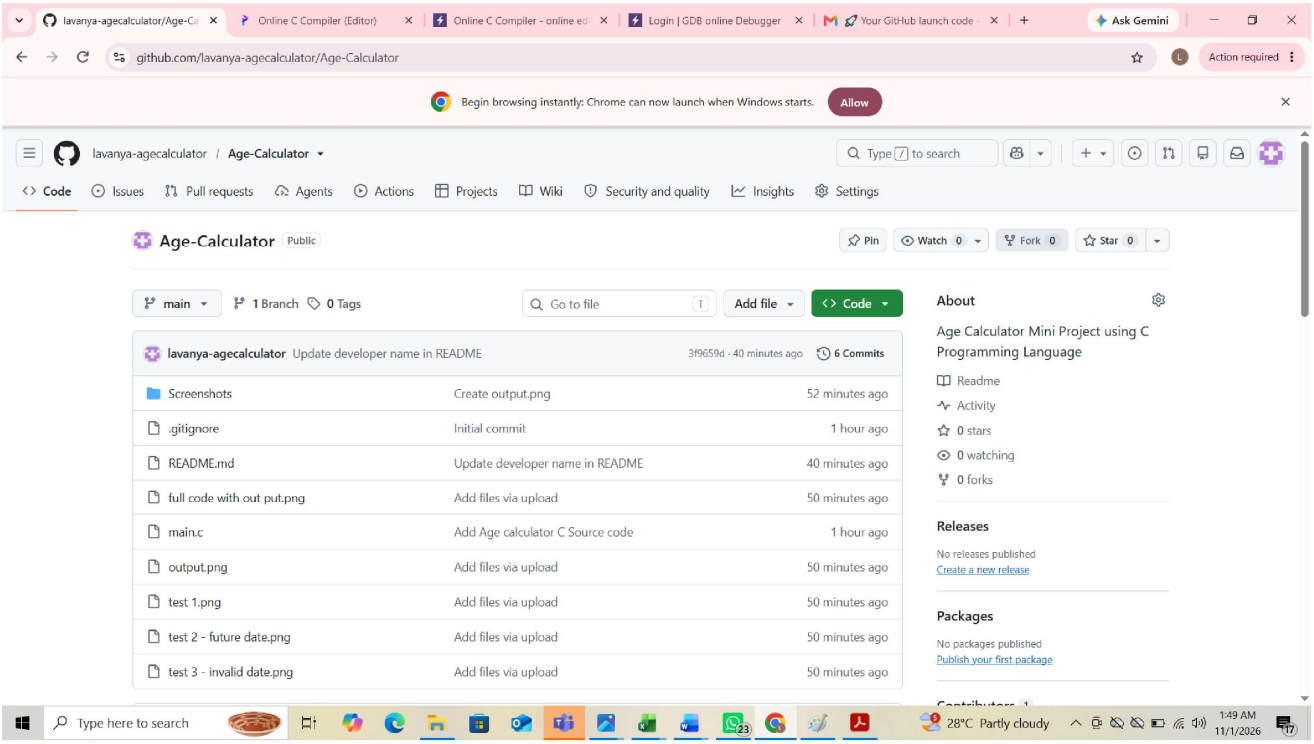
https://github.com/lavanya-agecalculator/Age-Calculator

6.6 Repository Structure

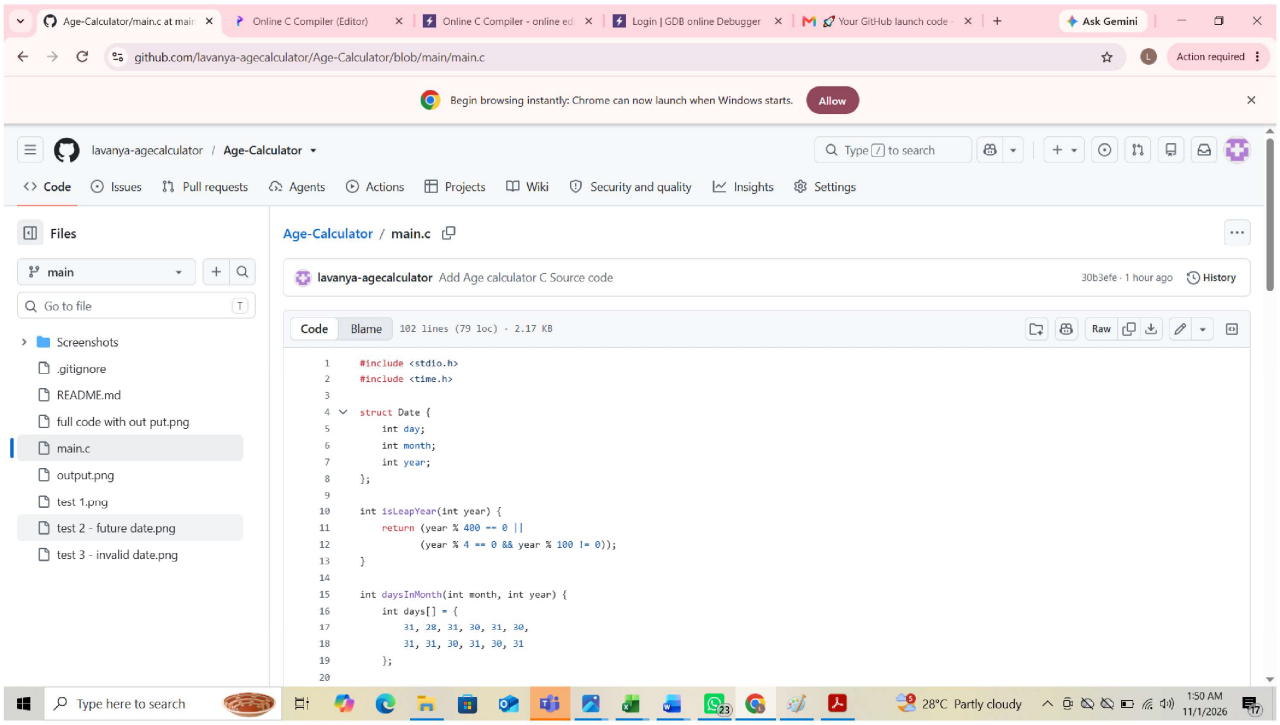
```
Age-Calculator/
├── main.c
├── README.md
├── .gitignore (if present)
└── Screenshots/ (if uploaded)
```

6.7 Screenshots of GitHub Repository

1. Public GitHub repository showing project files.



2. Main.c file displayed on GitHub.



Chapter 7 – Project Deployment

7.1 Introduction to Deployment

Deployment makes an application available for users to access. The current project is a standalone C console application.

7.2 Deployment Platform Used

A separate web deployment platform is not applicable because this version has no web interface. The source code is hosted in the GitHub repository.

7.3 Deployment Process

To run the project, open main.c in an online C compiler such as OnlineGDB, select C, compile/run the program and enter the date of birth.

7.4 Live Project URL

Not applicable for the current console-based version.

GitHub repository: https://github.com/lavanya-agecalculator/Age-Calculator

7.5 Screenshots of Deployment

An execution screenshot from the online compiler is included in Chapter 5.

Chapter 8 – Future Enhancements and Conclusion

8.1 Limitations of the Current Project

- The program is console-based and has no graphical user interface.
- It accepts one date of birth per execution.
- It does not save previous calculations or user records.
- The calculation uses the current system date/time supplied by the online compiler.

8.2 Future Enhancements

- Add a menu to calculate age for multiple users in one run.
- Develop a graphical or web-based interface.
- Add additional date-related calculations such as days until the next birthday.
- Store user records only with appropriate privacy and consent safeguards.

8.3 Conclusion

The Age Calculator mini project demonstrates the use of C programming fundamentals to accept and validate a date of birth and calculate age in years, months and days. The project helped practise structures, functions, conditional logic, arrays and date/time handling. The source code is maintained in a public GitHub repository.

References / Bibliography

23. OnlineGDB C Compiler, https://www.onlinegdb.com/online_c_compiler (accessed 01 November 2026).
24. GitHub Docs – Quickstart for repositories, https://docs.github.com/en/repositories/creating-and-managing-repositories/quickstart-for-repositories (accessed 01 November 2026).
25. GitHub Docs – Adding a file to a repository, https://docs.github.com/en/repositories/working-with-files/managing-files/adding-a-file-to-a-repository (accessed 01 November 2026).